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Subject name: Design Thinking and Idea Lab

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TECHNICAL REPORT

Title of the Project:IoT-Based Monitoring System for Solar/Wind Microgrids

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1. ABSTRACT

Renewable energy systems such as solar and wind microgrids are increasingly used in rural and remote areas to provide sustainable electricity. However, the lack of proper monitoring systems often leads to inefficiencies, battery failures, and high maintenance costs. This project proposes an **IoT-based monitoring system** that enables real-time monitoring of solar and wind microgrid parameters such as voltage, current, temperature, battery level, and power output. The system uses sensors connected to a microcontroller (ESP32) that collects data and sends it to a cloud platform through Wi-Fi or GSM. The collected data can be accessed through a dashboard for monitoring and analysis. This system helps detect faults early, improves system reliability, and reduces maintenance efforts. The proposed solution is cost-effective, scalable, and supports sustainable energy management in rural microgrid systems.

2. INTRODUCTION

Renewable energy sources like solar and wind are widely used for clean electricity generation. Many rural areas depend on microgrid systems for power supply. However, lack of proper monitoring can cause inefficiencies and system failures. IoT technology helps monitor these systems in real time using sensors and internet connectivity. It allows remote monitoring of voltage, current, and battery status. This improves system reliability and reduces maintenance cost. The proposed system provides smart monitoring for solar and wind microgrids.

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2.1 Project Overview

The The system monitors solar and wind microgrids using IoT technology. Sensors measure parameters such as voltage, current, temperature, and battery level. The collected data is processed using an ESP32 microcontroller. The system calculates power output and monitors battery efficiency. The data is then transmitted to a cloud platform through Wi-Fi or GSM. Users can view the system performance through a dashboard. If any abnormal condition such as low voltage or battery failure is detected, the system sends alerts and records the data for analysis.

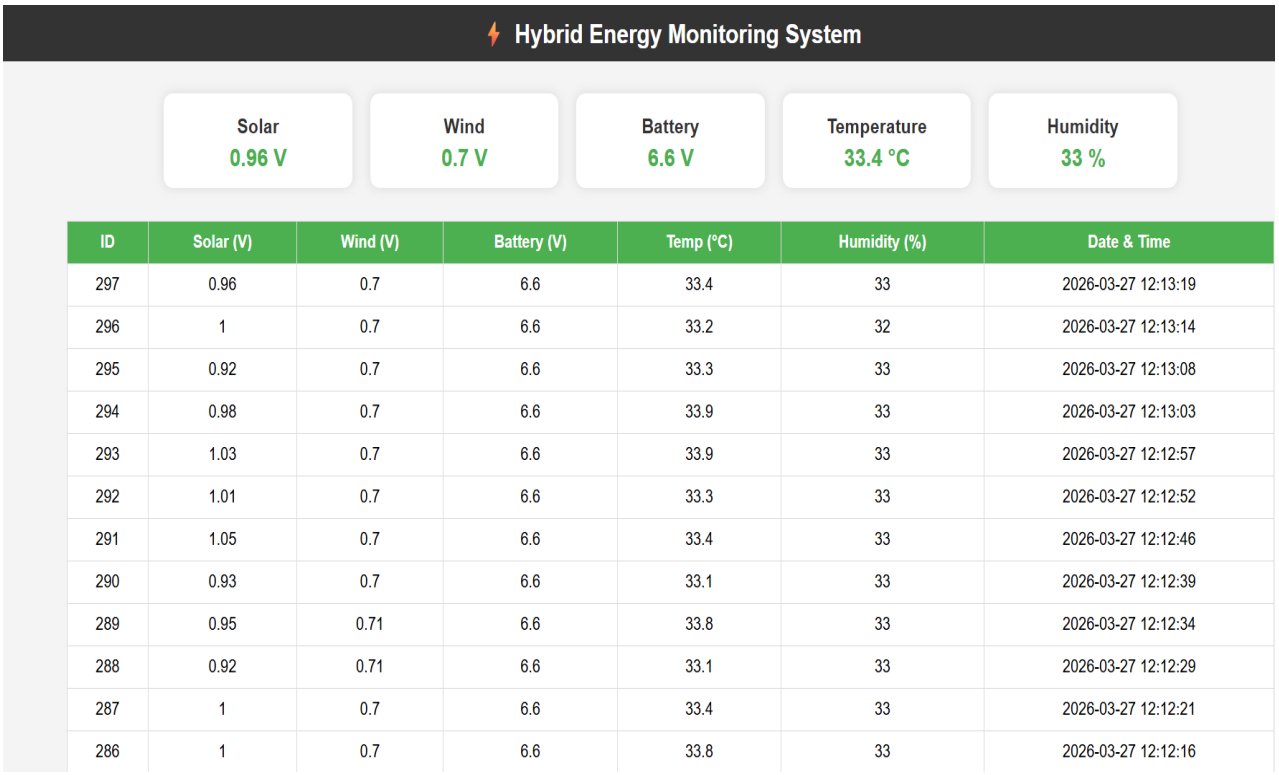
2.2 Objectives

- To monitor solar and wind microgrid parameters in real time
- To collect voltage, current, and temperature data using sensors
- To calculate power output and battery efficiency
- To send system data to a cloud platform for remote monitoring
- To detect faults early and improve system reliability

3. SYSTEM OVERVIEW

The proposed system is an IoT-based monitoring system for solar and wind microgrids designed to track the performance of renewable energy sources in real time. The system uses sensors to measure important parameters such as voltage, current, temperature, and battery level. These sensors are connected to an ESP32 microcontroller, which collects and processes the data.

The processed data is transmitted to a cloud platform through Wi-Fi or GSM communication. Users can access this information through a dashboard that displays the system status and performance. The system continuously monitors the microgrid and helps identify issues such as battery failure, low power generation, or abnormal temperature.



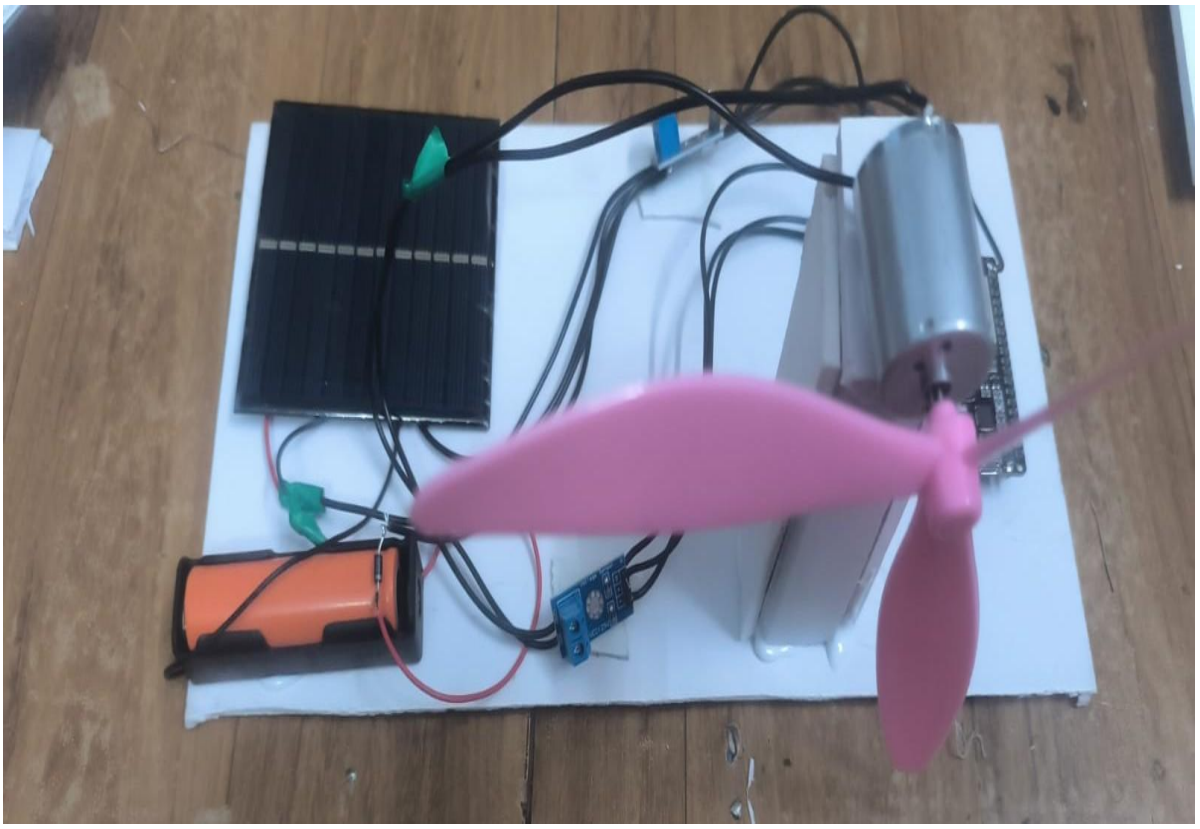
ID	Solar (V)	Wind (V)	Battery (V)	Temp (°C)	Humidity (%)	Date & Time
297	0.96	0.7	6.6	33.4	33	2026-03-27 12:13:19
296	1	0.7	6.6	33.2	32	2026-03-27 12:13:14
295	0.92	0.7	6.6	33.3	33	2026-03-27 12:13:08
294	0.98	0.7	6.6	33.9	33	2026-03-27 12:13:03
293	1.03	0.7	6.6	33.9	33	2026-03-27 12:12:57
292	1.01	0.7	6.6	33.3	33	2026-03-27 12:12:52
291	1.05	0.7	6.6	33.4	33	2026-03-27 12:12:46
290	0.93	0.7	6.6	33.1	33	2026-03-27 12:12:39
289	0.95	0.71	6.6	33.8	33	2026-03-27 12:12:34
288	0.92	0.71	6.6	33.1	33	2026-03-27 12:12:29
287	1	0.7	6.6	33.4	33	2026-03-27 12:12:21
286	1	0.7	6.6	33.8	33	2026-03-27 12:12:16

3.1 Microgrid monitoring hardware setup

The hardware setup consists of a solar panel, small wind turbine (DC motor with propeller), battery, and sensors mounted on a base board. The solar panel converts sunlight into electrical energy, while the wind turbine generates power from wind using a rotating propeller connected to a DC motor.

The generated power is stored in a rechargeable battery for later use. A voltage sensor module is used to measure the output voltage from the solar panel and wind turbine. The components are connected using wires and fixed on a stable board to form a small microgrid prototype.

This setup helps demonstrate how renewable energy sources can generate and store power while being monitored by an IoT-based system.



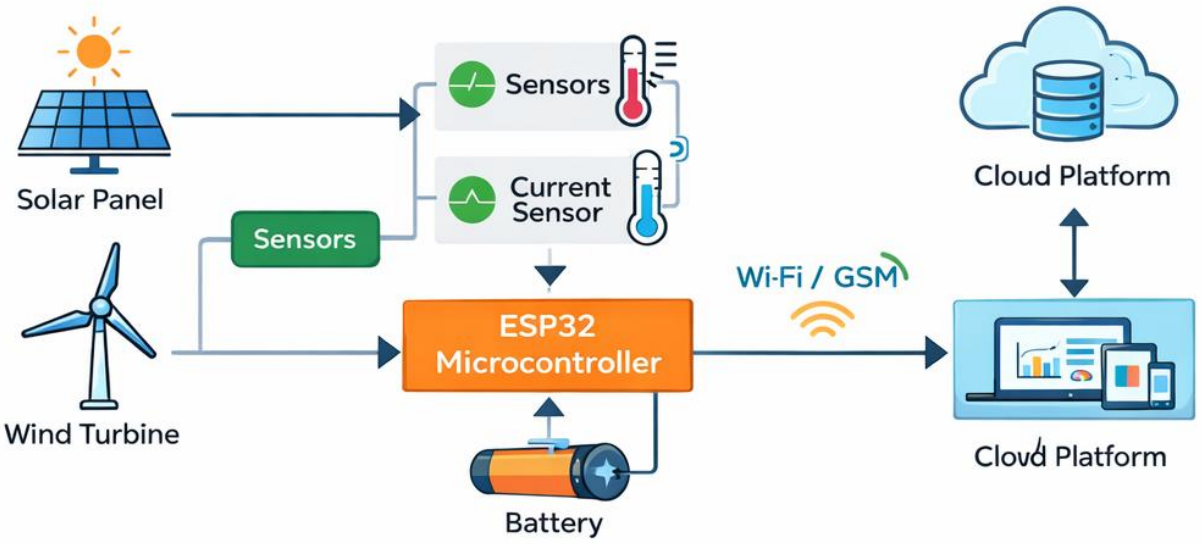
4. SYSTEM ARCHITECTURE

- The system consists of renewable energy sources, sensors, a microcontroller, and a cloud platform.
- Solar panel and wind turbine generate electrical energy.
- Voltage and current sensors measure electrical parameters.
- The ESP32 microcontroller collects and processes sensor data.
- Data is transmitted through Wi-Fi or GSM communication.
- The data is stored on a cloud server for monitoring and analysis.
- Users can access the system through a dashboard for real-time monitoring.

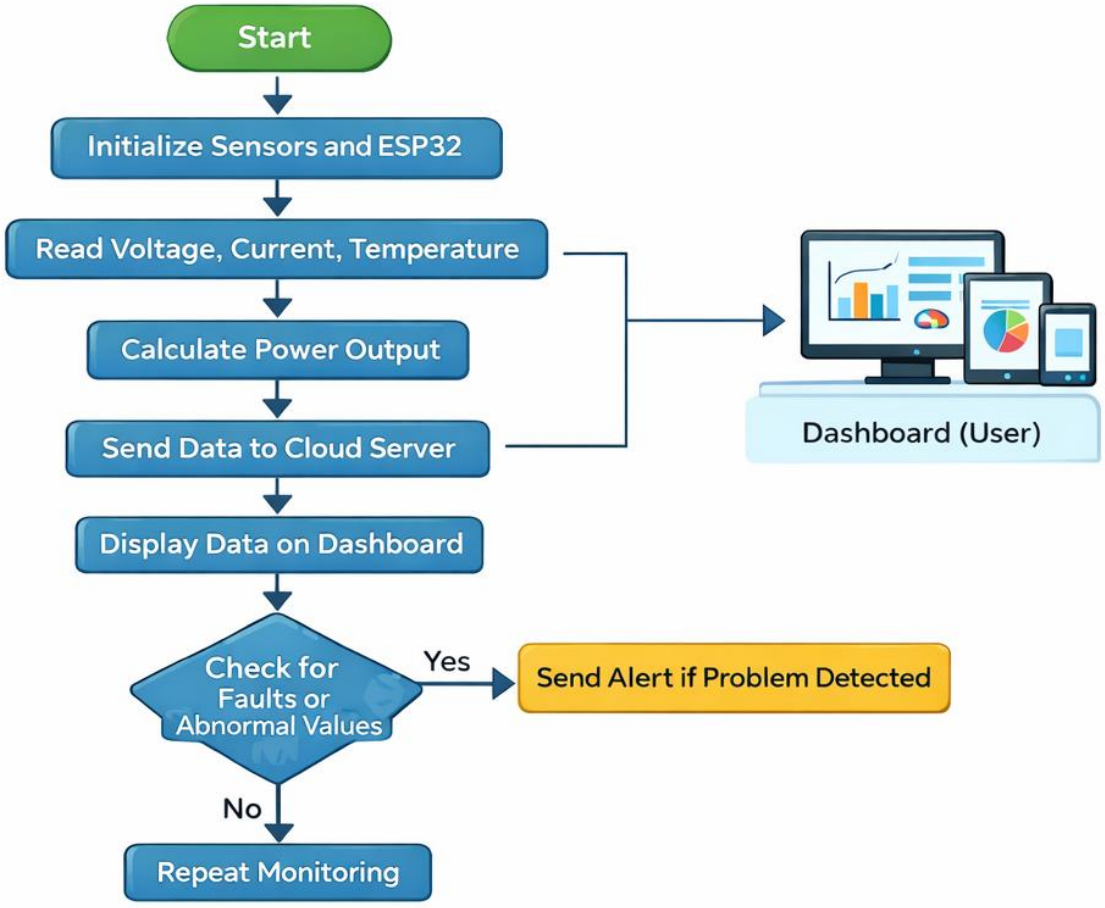
5.HARDWARE COMPONENTS

- **Solar Panel** – Converts sunlight into electrical energy.
- **Wind Turbine (DC Motor with Propeller)** – Generates power from wind energy.
- **ESP32 Microcontroller** – Main controller for data processing and communication.
- **Voltage Sensor** – Measures voltage produced by the system.
- **Current Sensor** – Measures current flow in the circuit.
- **Temperature Sensor** – Monitors system temperature.
- **Battery** – Stores generated energy for later use.
- **Charge Controller** – Regulates battery charging and prevents overcharging.
- **Wi-Fi / GSM Module** – Sends data to cloud platform.

Block Diagram



Flow Chart



- Start
- Initialize Sensors and ESP32
- Read Voltage, Current, Temperrature
- Hardware Components
- ESP32 Microcontrollec ESP32
- Wind Turbine DC Motor

5. SOFTWARE DESIGN

- The software is developed using Arduino IDE for ESP32 programming.
- The program reads data from voltage, current, and temperature sensors.
- Power is calculated using Power = Voltage × Current.
- Sensor data is transmitted to the cloud platform through Wi-Fi/GSM.
- The cloud platform stores data and displays it on a monitoring dashboard.

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6. WORKING PRINCIPLE

- Solar panel and wind turbine generate electrical power.
- Sensors measure **voltage, current, and temperature** continuously.
- The ESP32 microcontroller collects and processes the sensor data.
- The system calculates **power generation and battery efficiency**.
- Data is transmitted to the **cloud server for remote monitoring**.
- Users can monitor system performance through an **online dashboard**.
- Alerts are generated when abnormal conditions occur.

7. RESULTS AND ANALYSIS

- The system successfully monitors voltage, current, and power output.
- Real-time data is displayed on the monitoring dashboard.
- The system helps detect battery faults and low power generation early.
- Continuous monitoring improves system efficiency and reliability.
- Data analysis helps in better energy management.

8. APPLICATIONS AND ADVANTAGES

Applications

- Rural renewable energy monitoring
- Smart grid systems
- Solar power plant monitoring
- Wind energy monitoring systems

Advantages

- Real-time monitoring of renewable energy systems
- Early fault detection
- Reduced maintenance cost
- Remote monitoring capability

9. CODE AND CONCLUSION

Pseudocode

START
Initialize serial communication
Initialize DHT sensor

Connect to WiFi using SSID and Password
WHILE WiFi is not connected
    Wait for connection
END WHILE

LOOP forever

    Read analog value from Solar sensor pin
    Read analog value from Wind sensor pin
    Read analog value from Battery sensor pin

    Convert raw ADC values to actual voltages
        Solar Voltage = (Raw Solar / ADC Resolution) × Reference Voltage × Divider Ratio
        Wind Voltage = (Raw Wind / ADC Resolution) × Reference Voltage × Divider Ratio
        Battery Voltage = (Raw Battery / ADC Resolution) × Reference Voltage × Divider Ratio

    Read temperature from DHT sensor
    Read humidity from DHT sensor

    IF temperature OR humidity is invalid
        Display error message
        Skip current cycle
    END IF

Display Solar Voltage, Wind Voltage, Battery Voltage, Temperature and Humidity on serial monitor

IF WiFi is connected
Create URL with sensor data parameters
Send HTTP GET request to server
Receive server response
Display response on serial monitor
ELSE
Display connection error
END IF

Wait for 5 seconds

END LOOP

END

Conclusion

The proposed IoT-based monitoring system helps monitor solar and wind microgrid performance in real time. Sensors collect important parameters such as voltage, temperature, and battery status. The ESP32 microcontroller processes the data and sends it to a cloud server through Wi-Fi. This allows users to monitor the system remotely. The system helps detect faults early and improves energy management. Overall, it enhances the efficiency and reliability of renewable energy systems.